

TECHNICAL b REPORT

Plantigrade Robot

Chebyshev Linkage

Mechanical Engineering | Robotics | CAD

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Project Overview

Context

This project investigates the design and implementation of a small plantigrade walking mechanism based on a Chebyshev linkage.

Objectives

The main objectives are:

- Design the mechanism using CAD.
- Analyze its kinematic behavior.
- Select an appropriate actuator.
- Manufacture the mechanism using additive manufacturing.
- Validate the theoretical trajectory experimentally.

Engineering Requirements

The main technical requirements are summarized below.

Parameter	Target	Value	Unit
Motor speed	10–20	15	rpm
Supply voltage	6	6	V
Mechanism mass	< 500	420	g

Engineering Analysis

Motor Torque

The required torque can be approximated by:

$$\tau = I\alpha$$

where I is the moment of inertia and α is the angular acceleration.

DESIGN DECISION

A low-speed geared motor was selected to prioritize torque output over rotational speed.

Conclusion

The developed mechanism provides a first proof of concept for a compact plantigrade walking system.

Further development should focus on experimental validation, joint friction and optimization of the linkage geometry.